



Recommendation Systems with Movie Ratings

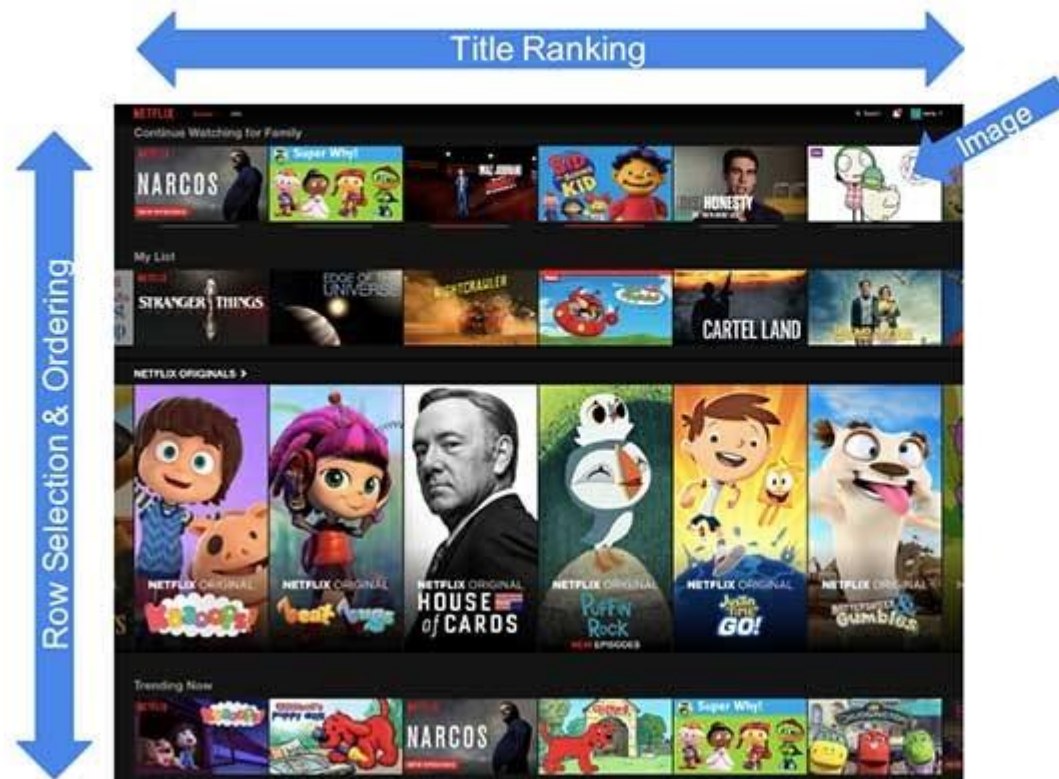
ECE 491

Ene Mecho



Introduction

Everything is a Recommendation



Recommendations are driven by machine learning algorithms

Over 80% of what members watch comes from our recommendations



Data Source



MovieLens

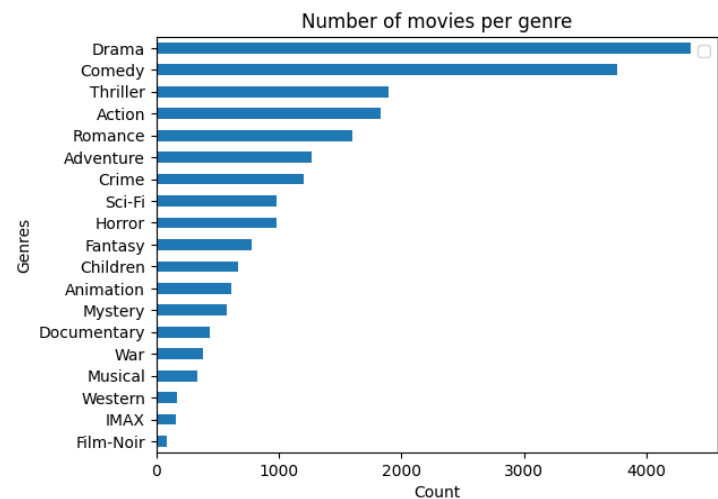
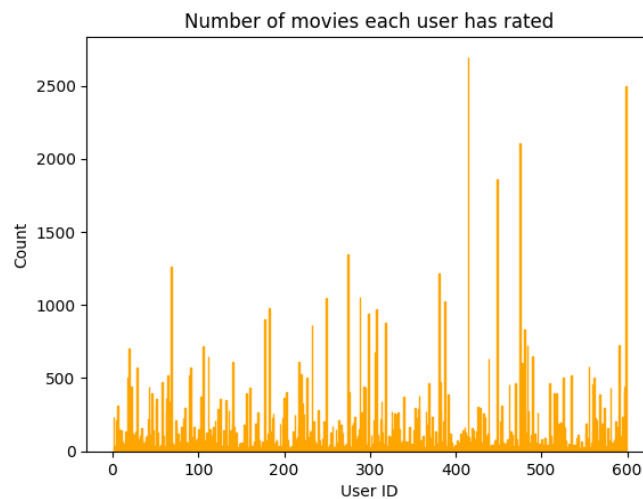
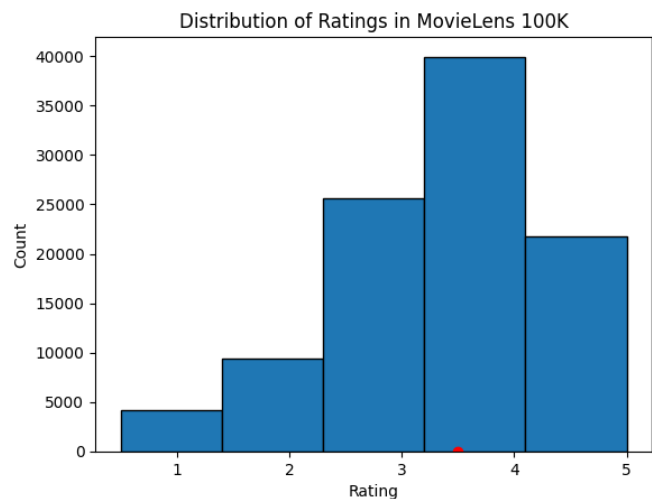


ml-latest-small

610 users
9,724 movies
100,863 ratings



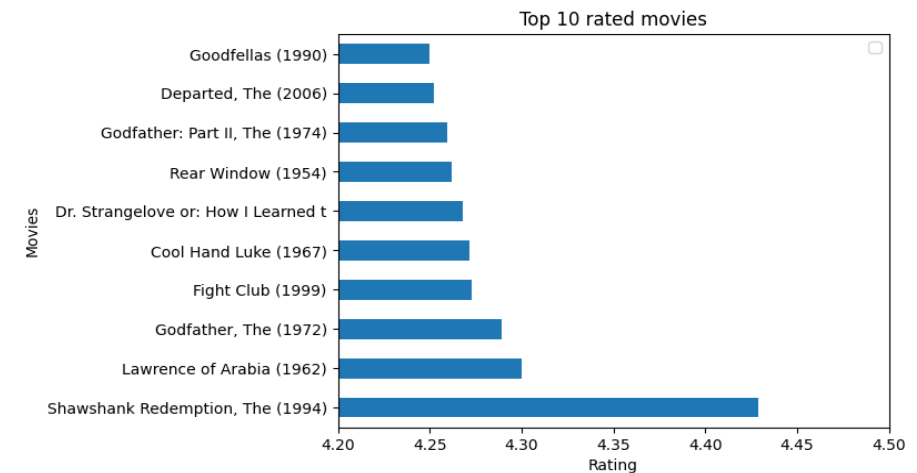
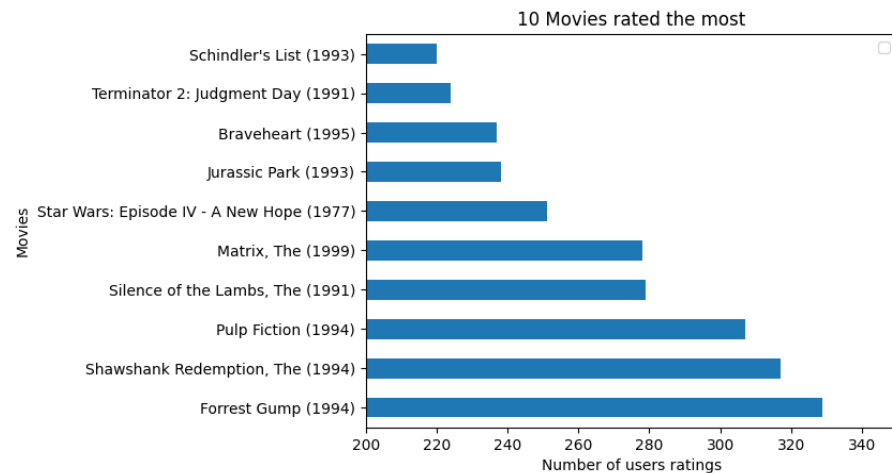
Exploratory Data Analysis



- 98.3% of the matrix is empty -> Sparse Matrix
- The mean of all ratings is 3.5 (left skewed distribution): Bias toward higher ratings
- Drama and Comedy are two top genres in this dataset
- There are some users that are outliers (Cinephiles)



EDA (cont.)



For the most part, the most popular movies aren't the best ranked

Note: Showing movies with more than 30 ratings in the "top 10 rated movies" plot



Setup

Majority of the setup is based on the size of the dataset:

- Train - Test ratio used: 90% - 10%
- No validation set (hyperparameters picked by inspection)
- Data split done randomly instead of sequence-aware



Popularity Baseline

Movie
"Rear Window"

User 1	2.5
User 2	3.5
User 3	5
User 4	3
User 5	4
User 6	?

$$\mu_j = \frac{\sum_{i=1}^{N_j} r_{ij}}{N_j}$$

r_{ij} : Rating of user i for the movie j
 N : Number of raters for the movie j
 μ_j : Average rating of the movie j

Steps

Calculate μ_j for every column j (movie j)



Autofilling every column j with the same μ_j



Masking already seen movies for every row j (user j)



Recommending the movies with the highest scores

Smoothed mean

$$\hat{\mu}_j = \frac{\sum_{i=1}^{N_j} r_{ij}}{N_j + 1}$$



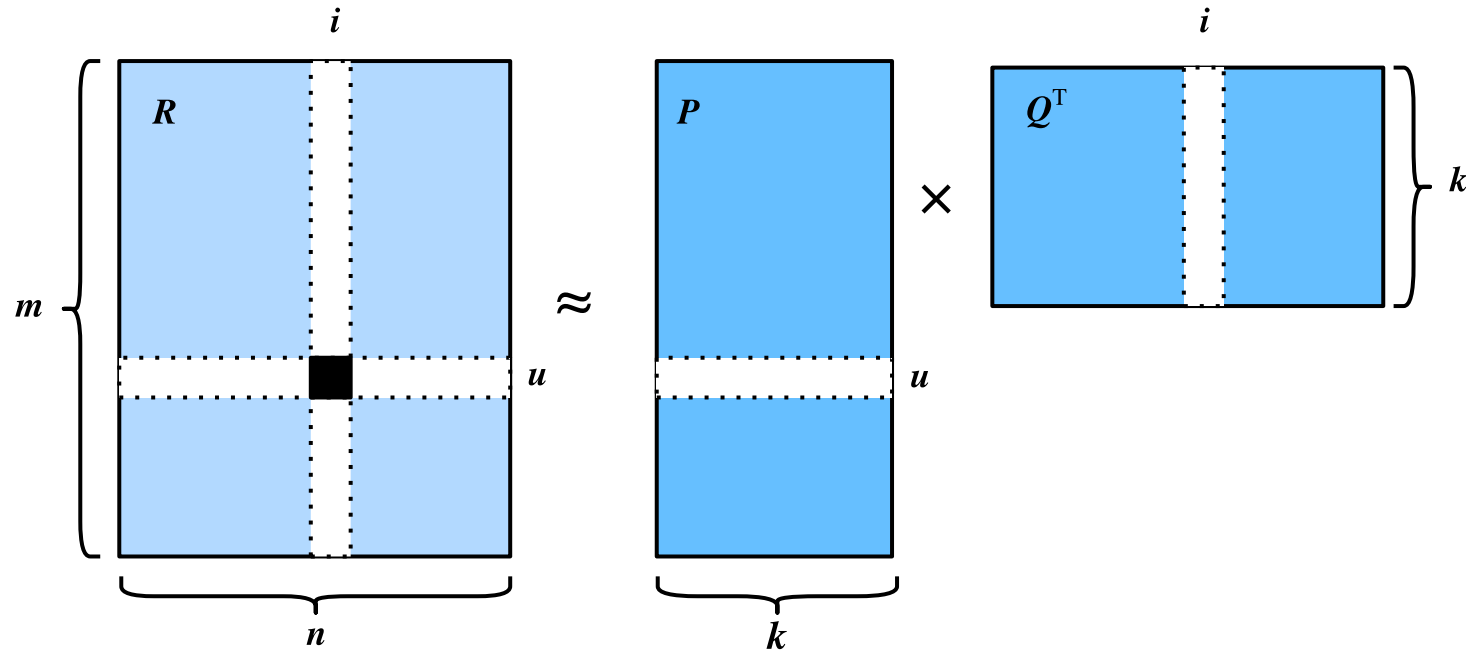
Popularity Baseline

	1	2	3	4	5	6	7	8	9	10
userId										
1	Shawshank Redemption, The (1994)	Casablanca (1942)	Streetcar Named Desire, A (1951)	Godfather: Part II, The (1974)	Secrets & Lies (1996)	Rear Window (1954)	Departed, The (2006)	Godfather, The (1972)	Dark Knight, The (2008)	Dr. Strangelove or: How I Learned to Stop Worr...
2	Casablanca (1942)	Streetcar Named Desire, A (1951)	Godfather: Part II, The (1974)	Fight Club (1999)	Usual Suspects, The (1995)	Goodfellas (1990)	Secrets & Lies (1996)	Rear Window (1954)	Godfather, The (1972)	Dr. Strangelove or: How I Learned to Stop Worr...
3	Shawshank Redemption, The (1994)	Casablanca (1942)	Streetcar Named Desire, A (1951)	Godfather: Part II, The (1974)	Fight Club (1999)	Usual Suspects, The (1995)	Goodfellas (1990)	Secrets & Lies (1996)	Rear Window (1954)	Departed, The (2006)
4	Shawshank Redemption, The (1994)	Streetcar Named Desire, A (1951)	Godfather: Part II, The (1974)	Usual Suspects, The (1995)	Secrets & Lies (1996)	Departed, The (2006)	Godfather, The (1972)	Dark Knight, The (2008)	Dr. Strangelove or: How I Learned to Stop Worr...	Lawrence of Arabia (1962)
5	Casablanca (1942)	Streetcar Named Desire, A (1951)	Godfather: Part II, The (1974)	Fight Club (1999)	Usual Suspects, The (1995)	Goodfellas (1990)	Secrets & Lies (1996)	Rear Window (1954)	Departed, The (2006)	Godfather, The (1972)

Pushes Conformity in movie taste (different users have different movie taste)



Matrix Factorization

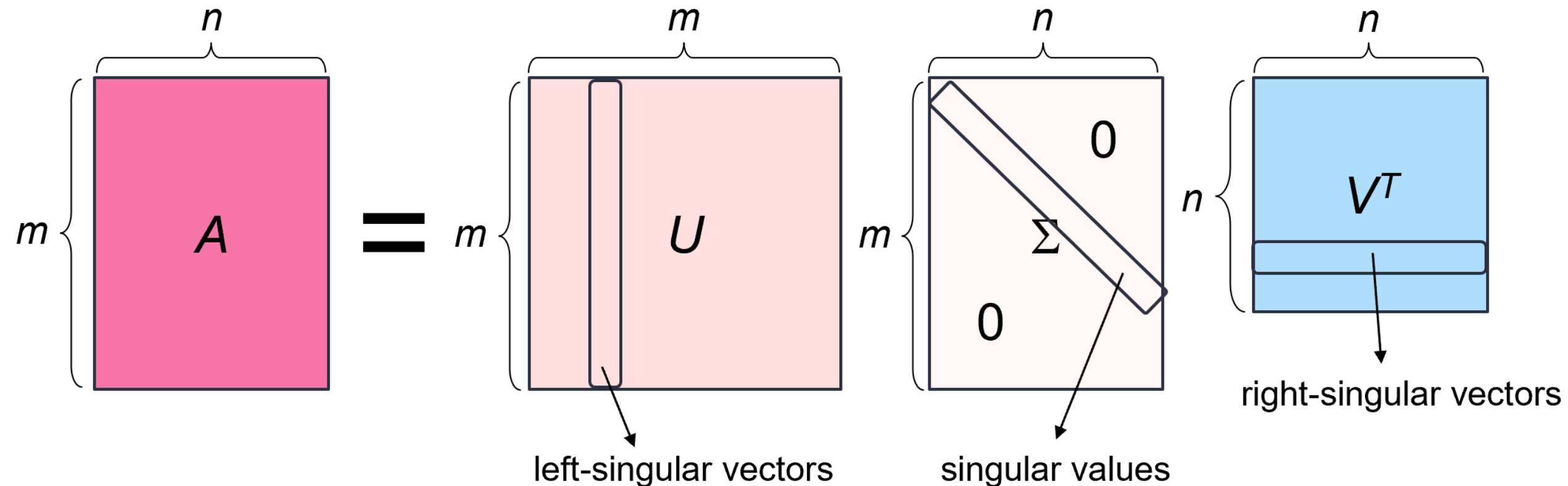


$$R = P \times Q^T$$

A matrix of m users and n movies can be decomposed into two matrices.
One matrix of m users with k latent variables, and the other matrix of n movies with k latent variables



Matrix Factorization (SVD)





Matrix Factorization (SVD)

	1	2	3	4	5	6	7	8	9	10
userId										
1	Raiders of the Lost Ark (Indiana Jones and the...	Blade Runner (1982)	2001: A Space Odyssey (1968)	Clockwork Orange, A (1971)	Twelve Monkeys (a.k.a. 12 Monkeys) (1995)	Shawshank Redemption, The (1994)	Lord of the Rings: The Fellowship of the Ring,...	O Brother, Where Art Thou? (2000)	Dr. Strangelove or: How I Learned to Stop Worr...	Eternal Sunshine of the Spotless Mind (2004)
2	Ace Ventura: Pet Detective (1994)	Dumb & Dumber (Dumb and Dumber) (1994)	Mask, The (1994)	Harry Potter and the Prisoner of Azkaban (2004)	Harry Potter and the Chamber of Secrets (2002)	Waterworld (1995)	Batman (1989)	Harry Potter and the Order of the Phoenix (2007)	Harry Potter and the Sorcerer's Stone (a.k.a. ...)	Harry Potter and the Goblet of Fire (2005)
3	Ace Ventura: Pet Detective (1994)	Ace Ventura: When Nature Calls (1995)	Mask, The (1994)	Starship Troopers (1997)	American Psycho (2000)	Predator (1987)	Matrix Reloaded, The (2003)	Face/Off (1997)	Independence Day (a.k.a. ID4) (1996)	Dumb & Dumber (Dumb and Dumber) (1994)
4	Back to the Future (1985)	Little Mermaid, The (1989)	Toy Story (1995)	2001: A Space Odyssey (1968)	Cinderella (1950)	E.T. the Extra-Terrestrial (1982)	Snow White and the Seven Dwarfs (1937)	Ghostbusters (a.k.a. Ghost Busters) (1984)	Toy Story 2 (1999)	To Kill a Mockingbird (1962)
5	Silence of the Lambs, The (1991)	Forrest Gump (1994)	Saving Private Ryan (1998)	Usual Suspects, The (1995)	Godfather, The (1972)	E.T. the Extra-Terrestrial (1982)	Godfather: Part II, The (1974)	Casablanca (1942)	Sixth Sense, The (1999)	Hunt for Red October, The (1990)

Customized for every user depending on their taste



Neural Recommender

Pearson Similarity

x_i are the individual values of one variable e.g. age

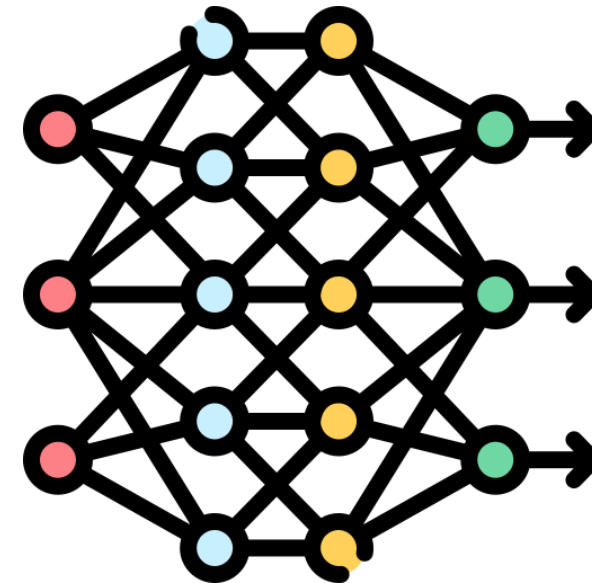
y_i are the individual values of the other variable e.g. salary

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Where r is the Pearson correlation coefficient,

$\bar{x}$ and $\bar{y}$ are respectively the mean values of the two variables.

Neural Network



+

MLP (3k->256->GELU->256->1)

Regularization Applied

k=32

3 channels concatenated (user, items, similarity matrix)



Neural Recommender

	1	2	3	4	5	6	7	8	9	10
userId										
1	Legend of Drunken Master, The (Jui kuen II) (1994)	Streetcar Named Desire, A (1951)	Manchurian Candidate, The (1962)	Hustler, The (1961)	Das Experiment (Experiment, The) (2001)	Logan (2017)	Top Secret! (1984)	Seventh Seal, The (Sjunde inseglet, Det) (1957)	Miller's Crossing (1990)	Right Stuff, The (1983)
2	Paths of Glory (1957)	Seventh Seal, The (Sjunde inseglet, Det) (1957)	Fear and Loathing in Las Vegas (1998)	Help, The (2011)	Big Sleep, The (1946)	Guess Who's Coming to Dinner (1967)	Lone Star (1996)	Gladiator (1992)	Last Detail, The (1973)	Brazil (1985)
3	Paths of Glory (1957)	Manchurian Candidate, The (1962)	Bottle Rocket (1996)	Braveheart (1995)	Gladiator (1992)	Brazil (1985)	Harold and Maude (1971)	Dial M for Murder (1954)	Third Man, The (1949)	Gallipoli (1981)
4	Back to the Future (1985)	Last Detail, The (1973)	Bound (1996)	On Golden Pond (1981)	Dawn of the Dead (1978)	Raging Bull (1980)	Inside Job (2010)	Unforgiven (1992)	Untitled Spider-Man Reboot (2017)	Persepolis (2007)
5	Manchurian Candidate, The (1962)	Secrets & Lies (1996)	Hustler, The (1961)	Gladiator (1992)	Harold and Maude (1971)	Streetcar Named Desire, A (1951)	Rear Window (1954)	Brazil (1985)	Infernal Affairs (Mou gaan dou) (2002)	Gallipoli (1981)

Including “niche” movies alongside popular ones makes recommendations much more specific



Results & Conclusion

Metrics	Popularity Baseline	Matrix Factorization	SVD M.F.	Custom Neural Recommender
RMSE	1.0313	0.9437	0.9050	0.8858
MAE	0.8083	0.7271	0.6988	0.6798

Conclusion: Popularity baseline is clearly the worst option, whereas my Neural Recommender got the highest metrics (RMSE & MAE). I would personally go with SVD since it's pretty accurate compared to the other two metrics while being less computationally intensive (no training).

RMSE: Root Mean Square Error (bigger penalty for bad guesses)
MAE: Mean Absolute Error (how many stars from the real rating we're off in average)



State-of-Art Comparison

“State-of-the-Art Recommender Systems” (April 2009)

	<i>Parameter</i>	<i>Learning time</i>	<i>prediction time</i>	<i>MAE</i>	<i>RMSE</i>	<i>precision₅</i>	<i>precision₄</i>
<i>model-based</i>	4 item clusters	5 sec.	1 sec.	0.6841	0.9172	0.5041	0.7550
<i>user-based</i>	300 neighbours	4 min.	3 sec.	0.6533	0.8902	0.5710	0.7810
<i>item-based</i>	100 neighbours	2 min.	1 sec.	0.6213	0.8550	0.5864	0.7915

Table 1: Summary of the best results on MovieLens depending on the type of approach.

	<i>Parameter</i>	<i>Learning time</i>	<i>prediction time</i>	<i>MAE</i>	<i>RMSE</i>	<i>precision₅</i>	<i>precision₄</i>
<i>model-based</i>	70 user clusters	24 min.	3 sec.	0.6566	0.8879	0.5777	0.7608
<i>user-based</i>	1,000 neighbours	9 h	28 min.	0.6440	0.8811	0.5902	0.7655
<i>item-based</i>	70 neighbours	2 h 30	1 min.	0.5990	0.8436	0.6216	0.7827

Table 2: Summary of the best results on Netflix depending on the type of approach.

"BellKor's Pragmatic Chaos" a Test RMSE of 0.8567 for Netflix Prize 2009

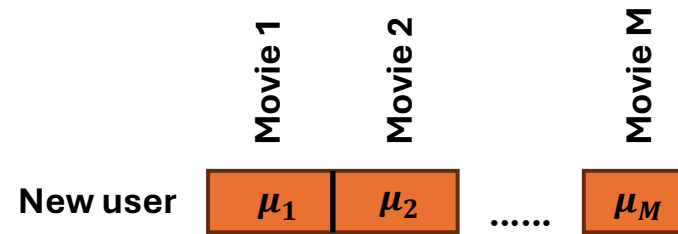
Different sized datasets

Different Results



Cold Start: New User

Using Popularity Baseline: $\mu_j = \frac{\sum_{i=1}^{N_j} r_{ij}}{N_j}$



Pipeline:

1. For every recommendation choosing the genre based on their distribution.
2. Recommending the highest rated movie for every genre.
3. If that genre is already included in the recommendation, recommend the second highest rated one and so on.
4. After having ranked 3 movies, we can insert him in the recommendation algorithms.



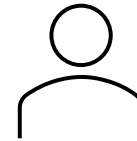
Cold Start: New Movie

“One battle after another”



Recommend?

User Ene

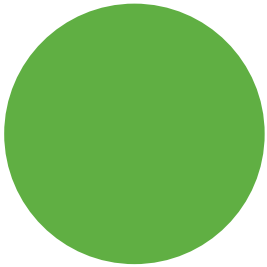


Trickier than the “New User” case

Approach:

1. Create a new recommendation section for New Movies
 2. Rank the new movies using the (smoothed) average rating of a user for a main genre
- Example: User Ene likes thrillers better. He gets new thriller movies recommended to him more than other genres.

Questions?





(Optional) CPU VS GPU Results

Model	RMSE	MAE
Matrix Factorization (k=10, Adam)	0.8718	0.6653
SVD Matrix Factorization (k=15)	0.8889	0.6860
Neural Recommender	0.8889	0.6847